In the case of large training data, a loss function $f : \mathbb{R}^n \to \mathbb{R}$ be of the form

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i(x), \qquad x \in \mathbb{R}^n. \tag{3.1}$$

Computing a full gradient

$$\nabla f(x) = \frac{1}{N} \sum_{i=1}^{N} \nabla f_i(x)$$

requires the evaluation of N individual gradients $\nabla f_i(x)$, which can be too expensive in case of using the gradient descent method. Instead, one could use the *stochastic gradient method* for minimizing $f : \mathbb{R}^n \to \mathbb{R}$ be: For given $x^0 \in \mathbb{R}^n$, iterate

(i) pick a random sample $d^k := d(x^k, \xi_k)$, where $\xi_k \sim \xi$ is distributed independently and identically. More precisely, $\xi_k = i(k)$ is distributed uniformly on $\{1, \dots, N\}$ and $d(x^k, \xi_k) = \nabla f_{i(k)}(x^k)$;

(ii) pick a step size $t_k > 0$;

(iii) set

$$x^{k+1} = x^k - t_k d^k.$$

The goal of this sheet is to show convergence results for stochastic gradient algoritm for smooth and convex functions under following assumptions:

(A1) $\mathbb{E}_\xi[d(x, \xi)] = \nabla f(x)$ for all $x \in \mathbb{R}^n$;

(A2) $\mathbb{E}_\xi\left[\|d(x, \xi)\|^2\right] \leq M^2$ for some $M > 0$ and all $x \in \mathbb{R}^n$.

Convergence is observed through the weighted running/ergodic mean

$$\bar{x}_\gamma^k := \sum_{j=0}^{k} \gamma_j x^j, \qquad \gamma_1, \dots, \gamma_k > 0$$

Exercise 11.1

Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth and μ-convex, and let $x^* \in \mathbb{R}^n$ be a minimizer of f.

1. Let the stepsize

$$t_k = \frac{1}{\mu(k+1)}$$

For the empirical mean $\bar{x}^k = \frac{1}{k+1}\sum_{j=0}^{k} x^j$, show that

$$\mathbb{E}\big[f(\bar{x}^k) - f(x^*)\big] \leq M^2 \frac{1+\log(k)}{2\mu(k+1)}.$$

Hint: Follow steps from 10.1 (including (A1) and (A2)) and Use strong convexity gradient inequality of f.

2. Let

$$t_k = t < \frac{1}{\mu},$$

Show that,

$$\mathbb{E}\big[\|x^k - x^*\|^2\big] - \frac{t}{\mu}M^2 \leq (1-t\mu)^k \left(\mathbb{E}\big[\|x^0 - x^*\|^2\big] - \frac{t}{\mu}M^2\right).$$

Interpret this result.

We now turn our attention to a different problem. One of the main challenges in modern data science is the enormous scale of contemporary optimization problems or large number of parameters to handle. This motivates the development of algorithms that rely only on partial gradient information. A classical approach in this direction is *Coordinate Descent*, where instead of updating the entire vector x^k at each iteration, only a single coordinate x_i^k is updated.

Exercise 11.2

Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and L-smooth. We consider a coordinate descent method given as

$$x^{k+1} = x^k - \frac{1}{L}\nabla_{i_k} f(x^k) e_{i_k}$$

for uniformly at random chosen $i_k \in \{1, \ldots, n\}$. (Here, $\nabla_j f(x^k) := \frac{\partial f(x^k)}{\partial x_j}$ and $e_j \in \mathbb{R}^n$ denotes the j's unit vector.) We assume that there exists a finite constant $R_0 > 0$ such that $\|x - x^*\| \leq R_0$ for any $x \in \mathbb{R}^n$ with $f(x) \leq f(x^0)$ and minimizer $x^* \in \mathbb{R}^n$ of f with $f(x^*) = f^*$.

1. Show that $(f(x^k))_{k\in\mathbb{N}}$ is monotonically decreasing by proving that

$$f(x^{k+1}) \leq f(x^k) - \frac{1}{2L}(\nabla_{i_k} f(x^k))^2.$$

2. Show that $\phi_k := \mathbb{E}[f(x^k)] - f^*$ fulfills

$$\phi_{k+1} \leq \phi_k - \frac{1}{2nL}(\mathbb{E}[\|\nabla f(x^k)\|])^2.$$

3. Use convexity of f to show that

$$f(x^k) - f^* \leq R_0 \|\nabla f(x^k)\|$$

and

$$\mathbb{E}[\|\nabla f(x^k)\|] \geq \frac{1}{R_0}\phi_k.$$

4. Show that $\frac{1}{\phi_{k+1}} - \frac{1}{\phi_k} \geq \frac{\phi_k - \phi_{k+1}}{\phi_k^2}$ for $k \in \mathbb{N}$

5. Prove that

$$\mathbb{E}[f(x^k)] - f^* \leq \frac{2nLR_0^2}{k}.$$